

- Q. 9.** Following table provides the details about 3 portfolios. Find out the difference between the actual and expected return using Jensen index. On the basis of your calculation rank them.

<i>Portfolio</i>	<i>Return on Portfolio</i>	<i>Portfolio Beta</i>	<i>Risk Free Interest Rate</i>
1	15	1.3	6%
2	12	0.9	6%
3	18	1.6	6%
Market Index	11	1.0	6%

Solution:

Jenson's Model: $R_p = \alpha + \beta (R_m - R_f)$

Portfolio 1: $R_{p1} = 6 + 1.3 (11 - 6)$

$$R_{p1} = 6 + 6.5$$

$$R_{p1} = 12.5$$

Portfolio 2: $R_{p2} = 6 + 0.9 (11 - 6)$

$$R_{p2} = 6 + 4.5$$

$$R_{p2} = 10.5$$

Portfolio 3: $R_{p3} = 6 + 1.6 (11 - 6)$

$$R_{p3} = 14$$

<i>Fund No.</i>	<i>Actual Return</i>	<i>Estimated Return</i>	<i>Difference</i>	<i>Rank</i>
1	15	12.5	2.5	I
2	12	10.5	1.5	II
3	18	14.0	- 4.0	III

- Q. 10.** With the given details, evaluate the performances of the different funds using Sharpe, Treynor and Jensen performance evaluation techniques.

<i>Funds</i>	<i>Return</i>	<i>Standard Deviation</i>	<i>Beta</i>
A	12	20	0.98
B	12	18	0.97
C	8	22	1.17
D	9	24	1.22

Risk free rate of return is 4 per cent. Market return is 10 %.

Solution: **Sharpe** $= \frac{R_p - R_f}{\sigma_p}$

Fund A: $= \frac{2-4}{20} = -0.1$... (4)

Fund B: $= \frac{12-4}{18} = 0.44$... (1)

Fund C: $= \frac{8-4}{22} = 0.18$... (3)

Fund D: $= \frac{4-4}{24} = 0.21$... (2)

Treyner $= \frac{R_p - R_f}{B_p}$

Fund A: $= \frac{2-4}{0.98} = -2.04$... (4)

Fund B: $= \frac{12-4}{.97} = 8.25$... (1)

Fund C: $= \frac{8-4}{1.17} = 3.42$... (3)

Fund D: $= \frac{9-4}{1.22} = 4.1$... (2)

Jensen:

$$R_p = \alpha + R_f + B(R_m - R_f)$$

Fund A: $2 = \alpha + 4 + .98 (10 - 4)$
 $\alpha = \mathbf{7.88}$... (1)

Fund B: $12 = \alpha + 4 + .97 (10 - 4)$
 $\alpha = 2.18$... (2)

Fund C: $8 = \alpha + 4 + 1.17 (10 - 4)$
 $\alpha = -\mathbf{3.02}$... (4)

Fund D: $9 = \alpha + 4 + 1.22 (10 - 4)$
 $\alpha = -\mathbf{2.32}$... (3)